

BELO HORIZONTE PAMPULHA, BrazilWMO: **835830**Lat: **19.851S**Lon: **43.951W**Elev: **789**StdP: **92.20**Time Zone: **-3.00 (SAB)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	12.1	3.8	5.5	25.3	5.1	6.0	24.0	7.1	21.4	6.4	21.4	0.8	100	0.285

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.9	33.1	19.6	32.1	19.8	31.1	19.9	22.6	28.3	22.2	27.9	21.8	27.3	2.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	17.5	23.7	21.0	17.2	23.5	20.2	16.4	23.3	71.0	28.6	69.3	27.7	67.7	27.6	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.4	5.7	5.2	DB	8.5	35.2	1.2	0.9	7.7	35.8	7.0	36.3	6.3	36.9	5.5	37.5	(4)
(5)			WB	6.9	24.0	1.3	1.5	6.0	25.1	5.2	25.9	4.5	26.7	3.5	27.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	22.7	24.9	25.3	24.3	22.9	20.8	19.6	19.4	20.6	22.7	23.9	23.6	24.2	(6)	
	DBStd	2.68	1.66	1.45	1.55	1.69	1.66	1.60	1.70	1.82	2.30	2.30	2.15	1.58	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	22	0	0	0	0	2	7	9	3	1	1	0	0	(9)	
	CDD10.0	4618	461	427	444	388	334	287	292	330	380	429	407	439	(10)	
	CDD18.3	1601	203	194	186	138	77	44	43	74	131	172	158	181	(11)	
	CDH23.3	11216	1528	1523	1234	826	433	301	338	598	1042	1276	991	1124	(12)	
(13)	CDH26.7	3406	544	557	381	179	57	28	38	142	366	460	312	341	(13)	
(14)	Wind	WSAvg	2.3	2.2	2.4	2.1	2.1	2.0	1.9	2.1	2.5	2.8	2.9	2.5	2.1	(14)
(15) Precipitation	PrecAvg	1453	294	156	171	64	29	11	4	10	47	101	230	335	(15)	
	PrecMax	2014	696	349	336	191	89	61	37	56	117	340	387	696	(16)	
	PrecMin	827	83	19	49	13	1	0	0	0	0	29	107	133	(17)	
	PrecStd	268	154	83	80	41	22	14	9	15	34	65	76	145	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.2	33.8	32.8	31.1	29.2	28.8	29.1	31.2	34.1	34.2	33.8	32.1	(19)	
		MCWB	20.4	21.7	21.2	20.9	19.2	17.6	16.9	16.7	17.9	18.4	20.2	20.3	(20)	
	2%	DB	32.0	32.2	31.1	29.9	28.1	27.2	27.8	29.9	32.1	32.8	31.2	31.0	(21)	
		MCWB	20.6	21.0	21.1	20.5	18.6	17.3	16.4	16.5	17.4	18.5	20.0	20.6	(22)	
	5%	DB	31.0	31.1	30.1	28.8	27.1	26.2	26.8	28.1	30.3	31.0	30.0	30.0	(23)	
		MCWB	20.5	20.7	20.9	20.2	18.4	17.1	16.3	16.1	17.5	18.7	19.9	20.4	(24)	
	10%	DB	29.9	30.1	29.0	27.8	26.0	25.1	25.2	26.9	28.9	29.2	28.2	28.9	(25)	
		MCWB	20.4	20.4	20.8	19.9	18.2	16.8	16.2	16.1	17.6	18.9	19.8	20.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.1	23.2	23.0	22.3	21.1	19.9	18.7	19.1	21.1	21.7	22.2	22.9	(27)	
		MCDB	28.3	29.6	28.7	27.6	26.3	25.9	25.4	26.4	28.0	27.3	28.0	28.0	(28)	
	2%	WB	22.4	22.4	22.2	21.7	20.2	18.8	18.0	18.4	20.1	21.0	21.6	22.2	(29)	
		MCDB	27.9	28.9	28.1	27.1	25.5	24.5	24.9	25.7	26.7	26.9	27.0	27.1	(30)	
	5%	WB	21.9	21.9	21.8	21.1	19.6	18.2	17.4	17.6	19.3	20.5	21.1	21.7	(31)	
		MCDB	27.4	28.2	27.6	26.3	24.8	23.8	24.2	24.9	25.7	26.4	26.1	26.4	(32)	
	10%	WB	21.4	21.5	21.4	20.6	18.9	17.5	16.9	17.0	18.7	20.1	20.7	21.3	(33)	
		MCDB	26.5	27.2	26.6	25.6	23.8	23.0	23.4	24.1	25.2	26.0	25.5	25.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	9.9	9.1	9.9	10.6	11.8	12.3	12.5	11.5	10.2	8.7	8.7	(35)	
		MCDBR	11.3	11.4	10.9	11.0	12.1	13.3	14.3	14.8	14.1	12.7	11.3	10.9	(36)	
	5% WB	MCWBR	3.3	3.2	3.4	3.8	4.4	5.0	5.0	4.5	3.9	3.7	3.4	3.4	(37)	
		MCWBR	9.3	9.9	9.3	9.7	10.3	11.6	11.9	12.1	10.6	10.1	9.0	8.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.392	0.383	0.378	0.357	0.338	0.326	0.330	0.362	0.446	0.440	0.403	0.396	(40)	
		taud	2.449	2.461	2.497	2.516	2.539	2.571	2.534	2.424	2.183	2.254	2.395	2.436	(41)	
	Ebn at Noon		953	950	931	911	893	886	892	893	852	886	936	949	(42)	
		Edh at Noon	122	119	110	101	91	85	90	108	148	144	128	123	(43)	
(44) All-Sky Solar Radiation	RadAvg		5.83	5.98	5.27	4.86	4.22	3.98	4.25	4.98	5.31	5.59	5.15	5.46	(44)	
	RadStd		0.86	0.65	0.47	0.31	0.24	0.26	0.24	0.33	0.54	0.65	0.48	0.57	(45)	

Historical Trends

	DBAvg	Heating
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